

SWGShield

Building a Web Security Monitoring, Protection, Detection, and Prevention System Integrated
with AI-based Scam Content Analysis

STUDENT	SUPERVISOR	SCHOOL	YEAR
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01 / The Problem

Traditional WAF Limitations

Standard tools struggle against modern threats.

- **Rule-based & Blacklist:** Old rules cannot catch newly formulated attacks.
- **Zero-day Blind Spot:** Incapable of detecting new attack signatures.
- **Lacks Semantic Context:** Easily bypassed by simple obfuscation (e.g., using "cllck" instead of "click").
- **No Continuous Learning:** Static rulesets degrade over time.

The AI Security Challenge

While Deep Learning offers superior accuracy in detecting sophisticated phishing and social engineering semantics, it introduces a critical flaw:

- High computational latency.
- Not suitable for real-time gateway deployments.
- Resource-intensive on massive traffic loads.

The Core Challenge: Balancing the deep understanding of AI with the millisecond-latency requirements of a web gateway.

| 02 / Objectives & Contributions



Multi-Layer AI Pipeline

Combines traditional WAF + QuickText (outer loop for ms-latency) with a Smart Checker (inner loop for semantic depth). Achieving both speed and correctness.



Explainable AI (LIME)

Moving beyond black-box Yes/No decisions. LIME integration highlights the exact trick words and manipulations, making the security verdict fully transparent.



Continuous Learning

Implemented a Human-In-The-Loop (HITL) system to verify unclear cases. Allows the model to update and hot-reload new versions with zero downtime.

03 / System Architecture

Request → Check → Smart Check → Save

Browser Extension: Chrome MV3 · Content Script

OWASP WAF: CRS v4.0 · SQLi / XSS Block

Gateway Orchestrator: QuickAPI :8000 · Rate Limit

QuickText (L2): :5001 · N-gram 3-8 · 0.03ms Latency

Smart Checker (L3): :5002 · Self-Attention GPU T4

XAI & DB: LIME Feature Importance · SQLite HITL

queue

| 04 / Defense Layer 1: WAF & Gateway

Web Application Firewall

- Utilizes standard `OWASP CRS v4.0` rules.
- Blocks known SQL Injection and XSS vulnerabilities right at the door.
- Inspects all headers and payloads before they ever reach the AI layer.

Rate Limiting & Anti-DDoS

- **Token Bucket Algorithm:** Enforces an exact limit of 60 req/min (1 token/sec).
- Automatically drops excess traffic, returning HTTP 429 status.
- Ensures system availability under flood attacks.

 Abstract glowing digital shield representing firewall and web security

| 05 / Defense Layer 2: Quick Check

0.03_{ms}

INFERENCE LATENCY

98.5%

ACCURACY RATE

>30k

REQUESTS / SECOND

High-Speed Heuristic Shield

QuickText operates as a lightweight outer loop utilizing Sub-Word N-grams (3-8 characters). This effectively counters deliberate typos and out-of-vocabulary obfuscation (e.g., `` tokenized to vectors).

Routing Logic

- ✓ If Confidence > 95% Safe → Bypass immediately.
- ⚠ If Suspicious / Edge Case → Route to Smart Checker (L3).

Maintains only ~18 MB RAM footprint.

06 / Defense Layer 3: Smart Understanding

Transformer Self-Attention

For complex payloads bypassed by L2, the Smart Checker applies deep Semantic NLP.

- Understands global context via Self-Attention, not just isolated terms.
- Optimized for GPU T4, retaining >95% capability of the original BERT model.
- Processes Max Sequence Length of 512 tokens instantly.
- Highly effective at catching sophisticated phishing narratives and semantic manipulation.

LIME: Explainable AI

AI decisions are no longer a black box. LIME (Local Interpretable Model-agnostic Explanations) breaks down the verdict.

```
> "You won a prize 1 billion VND.  
Click now to claim..."
```

click here : 0.94

ngay bây giờ: 0.82

Click now : 0.67

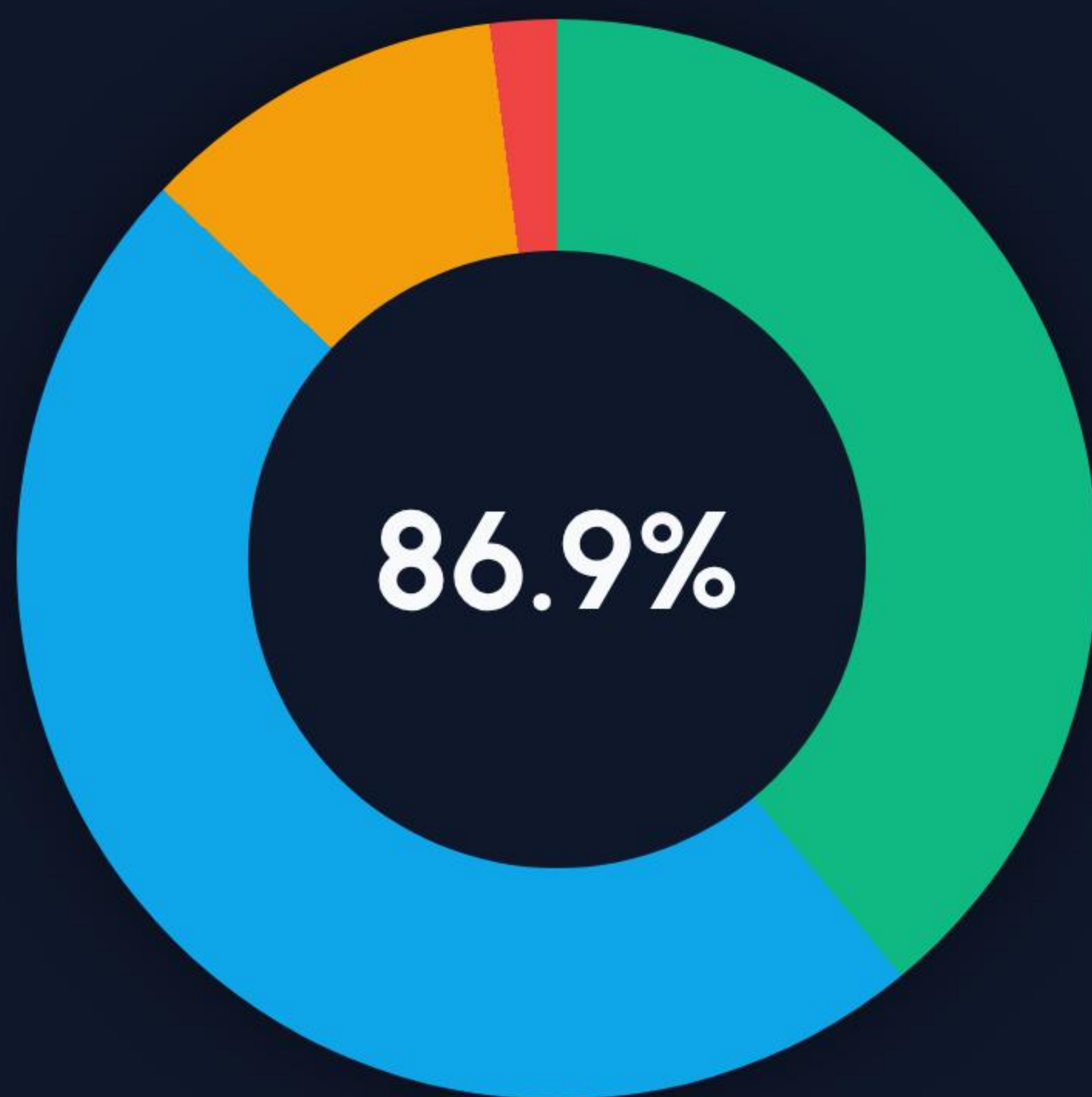


Verdict: PHISHING (Conf: 0.94)

| 07 / Continuous Learning Pipeline



08 / Results & Evaluation

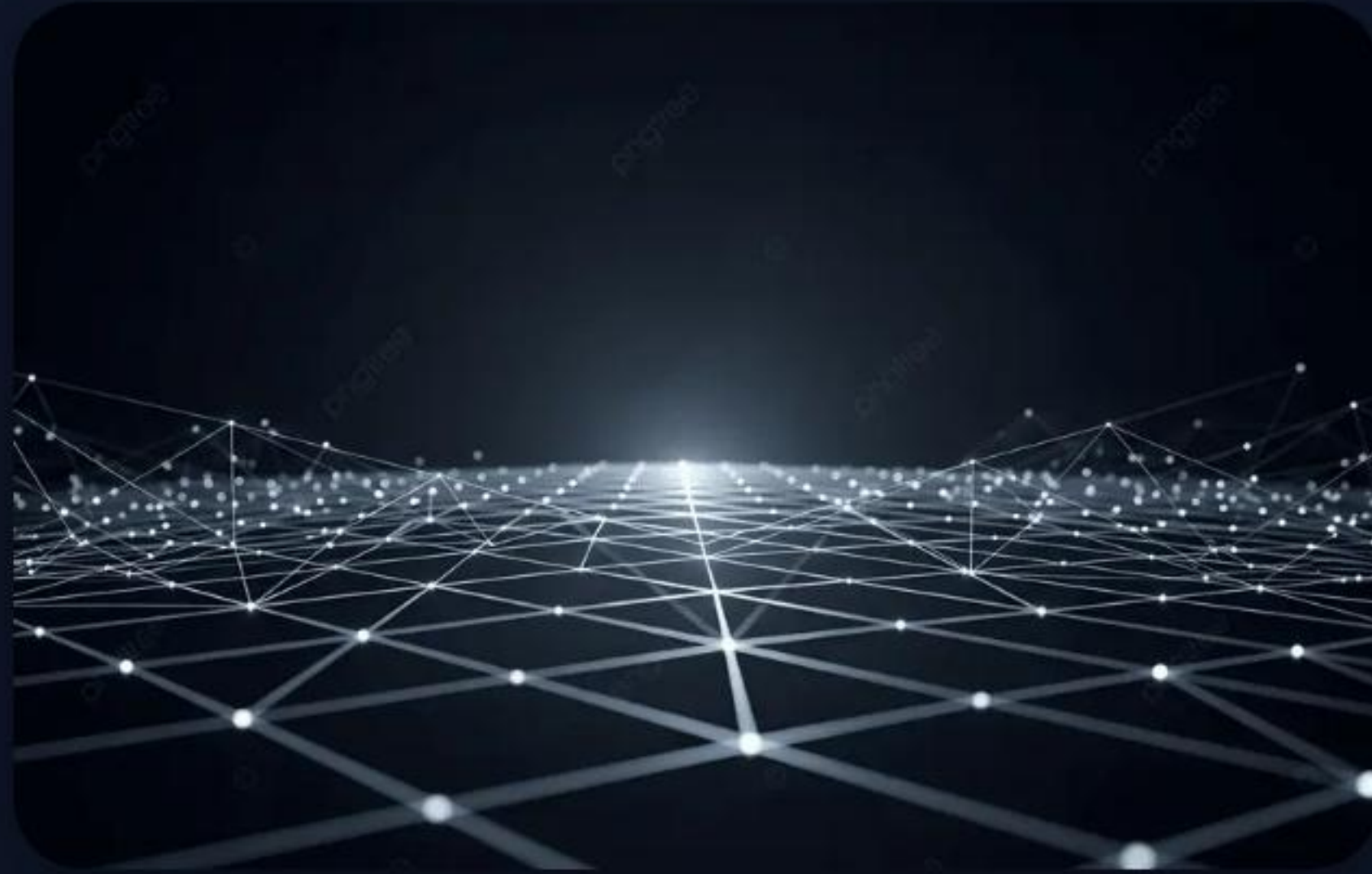


Test Set Performance

- True Positive (Scam caught): 9,243
- True Negative (Safe passed): 11,482
- False Negative (Missed scam): 2,679
- False Positive (Safe blocked): 440

Maintained low False Positive rate while securing 86.9% global accuracy on the strict validation subset.

09 / Conclusion & Future Roadmap



Vietnamese Dataset Expansion

Grow the localized Fraud Dataset to aggressively target Zalo scams, fake banking SMS, and OTP manipulation tactics.



Computer Vision Integration

Expand beyond text parsing to detect malicious payloads hidden inside images and fraudulent QR code injection.

 Abstract glowing 3d blocks shipping containers digital representation docker

Production Containerization

Full deployment pipeline transitioning to Docker for consistent environments and optimized cluster scaling.

Thank You!

Capstone Project — Secure Web Gateway System

[Q & A SESSION]

FOLLOWED BY LIVE SYSTEM DEMO

Image Sources



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